LCMS + GLOBE Observer Adopt-a-Pixel Workshop
NASA / USFS Public Facilitator Guide

This facilitator packet supports delivery of a public-facing LCMS Adopt-a-Pixel workshop integrated with GLOBE Observer land cover observations and imagery. It aligns with NASA Earth Science Education and U.S. Forest Service training standards.

1. Workshop Overview

The Adopt-a-Pixel activity introduces participants to satellite-derived land cover and landscape change products from the Landscape Change Monitoring System (LCMS), while using GLOBE Observer ground observations as contextual reference information.

2. Learning Objectives

By the end of this activity, participants will be able to:
• Interpret LCMS land cover, change year, and disturbance type at the pixel scale.
• Compare satellite-derived classifications with human-reported observations.
• Discuss spatial scale, temporal mismatch, and uncertainty in land cover mapping.

3. Data Sources

LCMS: Model-based estimates of land cover and landscape change at 30-meter resolution, derived from Landsat imagery.

GLOBE Observer: Volunteer-submitted land cover observations and photos collected via the GLOBE Observer mobile application.

4. Facilitation Guide

1. Ask participants to open the public LCMS + GLOBE Observer web application.
2. Instruct participants to adopt a single LCMS pixel by clicking on the map.
3. Toggle LCMS Land Cover, Change Year, and Disturbance Type layers.
4. Enable GLOBE Observer points and adjust buffer size to explore spatial context.
5. Click a GLOBE point to view associated ground photos alongside LCMS pixel information.

5. Discussion Prompts

• Do LCMS and GLOBE Observer observations agree?
• How does buffer size affect interpretation?
• What factors might explain disagreement between datasets?

6. Accessibility Considerations

The application uses high-contrast symbology, text-based legends, keyboard-accessible controls, and screen-reader-compatible panels. Images are optional and accompanied by explanatory text.

7. Important Disclaimers

LCMS products represent modeled estimates, not ground truth. GLOBE Observer data and images are used for interpretation and discussion only and are not validation data or accuracy assessments.